# Stemming VS Morphological Parsing

## DSA0305 - NATURAL LANGUAGE PROCESSING


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01

# Stemming

*A crude, rule-based chop*

```
studies  →  studi
studying  →  study
national → nation
```

## Definition

Stemming strips prefixes and suffixes from a word using heuristic rules, producing a “stem” that may not even be a real word — fast, but linguistically crude.

### How Stemming Works — the Porter algorithm applies ordered suffix-removal rules

1. **Tokenize**
   Split text into word tokens
2. **Match suffix rules**
   e.g., SSES→SS, IES→I
3. **Ordered passes**
   Rule sets erode the ending
4. **Return the stem**
   No dictionary check applied

Example pass: **relational** (ATIONAL→ATE) → **relate**

02

# Morphological Parsing

*A structured, linguistic analysis*

## Definition

Morphological parsing decomposes a word into its constituent morphemes (root + affixes) and labels their grammatical function — slower, but linguistically precise.

### How Morphological Parsing Works — a finite-state transducer maps form to structure

1. **Segment morphemes**
   Split into root + affixes
2. **Consult lexicon**
   Verify root & category
3. **Apply morphotactics**
   Confirm legal affix order
4. **Emit tagged lemma**
   Root + grammatical tags

Example parse: **unhappiness** → **happy + NEG(un-) + NMLZ(-ness)**

```
studies  →  study + NOUN + PL
running  →  run + V + PROG
kinder  →  kind + COMP
```

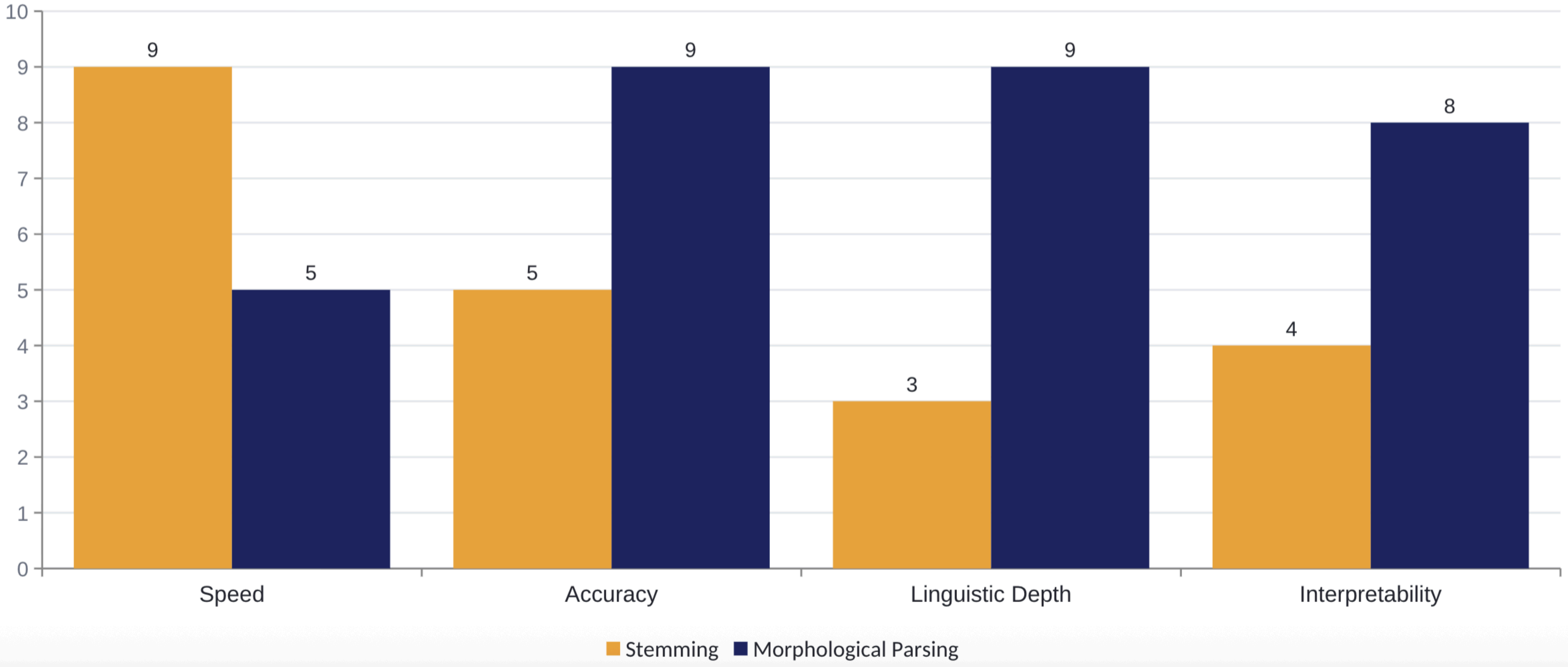
# Same Word, Two Outcomes

Tracing three input words through both pipelines

| INPUT | STEMMING OUTPUT | MORPHOLOGICAL PARSING OUTPUT |
|---|---|---|
| universities | `univers` | `university + NOUN + PL` |
| better | `better` | `good + ADJ + COMPAR` |
| argued | `argu` | `argue + VERB + PAST` |

# Comparative Analysis

Illustrative scores across four evaluation dimensions (0–10 scale)



# Side-by-Side Summary

| Dimension | Stemming | Morphological Parsing |
| --- | --- | --- |
| **Approach** | Heuristic suffix-stripping | Rule/lexicon-based morpheme analysis |
| **Output validity** | May produce non-words | Always a valid lemma + tags |
| **Speed** | Very fast | Slower, resource-dependent |
| **Linguistic resources** | None required | Lexicon + grammar rules required |
| **Typical use** | Search engines, information retrieval | Machine translation, grammar checking, TTS |

# Key Takeaways

Match the technique to what your task actually needs

### Choose Stemming when...

- Speed and scale matter more than precision
- Building search or IR indexes
- Linguistic resources are limited

### Choose Morphological Parsing when...

- Grammatical accuracy is essential
- You need POS or tense information
- Output must be valid, readable words

# THANK YOU!

Any Questions?